Using xG and xT to Construct a RAPM Model for the AHL

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An article presented in conjunction with the
36-660: Sports Analytics final project.

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April 15, 2026

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1 Abstract

2 Introduction

In hockey, the players who score the most goals are often the ones who win awards, sign the biggest contracts, and receive the most attention. However, in building a roster, it is important to look beyond raw box score statistics in order to better understand player performance and who actually contributes to winning. This is even more true for front offices analyzing performance in the American Hockey League (AHL), the best developmental league in the pipeline to the National Hockey League (NHL). Thus, front offices must use AHL results to determine who is worthy of an NHL promotion and project how those players will perform. Again, while the headlines generally revolve around big-name free agents or blockbuster trades, the teams that take advantage of their internal development pipeline are always the ones that experience long-term success. In this paper, we use Regularized Adjusted Plus Minuses (RAPMs) to analyze player effects on Expected Threat (xT) and Expected Goals (xG) in the 2023-24 AHL season. We finish with an investigation into how xT influence predicts NHL-level ability through promotion.

3 Data

3.1 Data Description

We used data from the 2026 Hockey Analytics League Operations (HALO) Hackathon. This data contains event-level tracking data (i.e. player tracking data given as a snapshot at the time of every event, but not continuously) from 480 games from the 2023-24 AHL season. There are approximately 1.8 million total events in the dataset, and approximately 1.35 million of these have associated tracking data. Each event contains contextual information, such as a game ID, the time of the event within the game, the location of the event on the ice, the type of event, whether it was successful or not, and some descriptive flags (e.g. "defensive zone outlet pass"). There are 1172 total players in the dataset.

In this paper, we frequently reference xT and xG. The data originally created for the HALO competition does contain xG for shots based on a model created by SportLogiq, but we did not use this model for the xG-based part of our project. This is because we used the xT metrics created by Michele Sezgin and Tyler Quinn for the HALO 2026 competition (Sezgin & Quinn, 2026). In that project, they also created an xG model that was used in the creation of their xT metric, so we are using the xG values on shots from that model in the interest of consistency.

We also downloaded data from Hockey Reference on every NHL player over the last three seasons (2023-24, 2024-25, and 2025-26). The purpose of this data is to determine which players in the AHL dataset made it to the NHL during or after the time the AHL data was collected.

3.2 Exploratory Data Analysis

First, we examined the distributions of shift-level net xG and xT by period in order to see if differences in time needed to be addressed in our models.

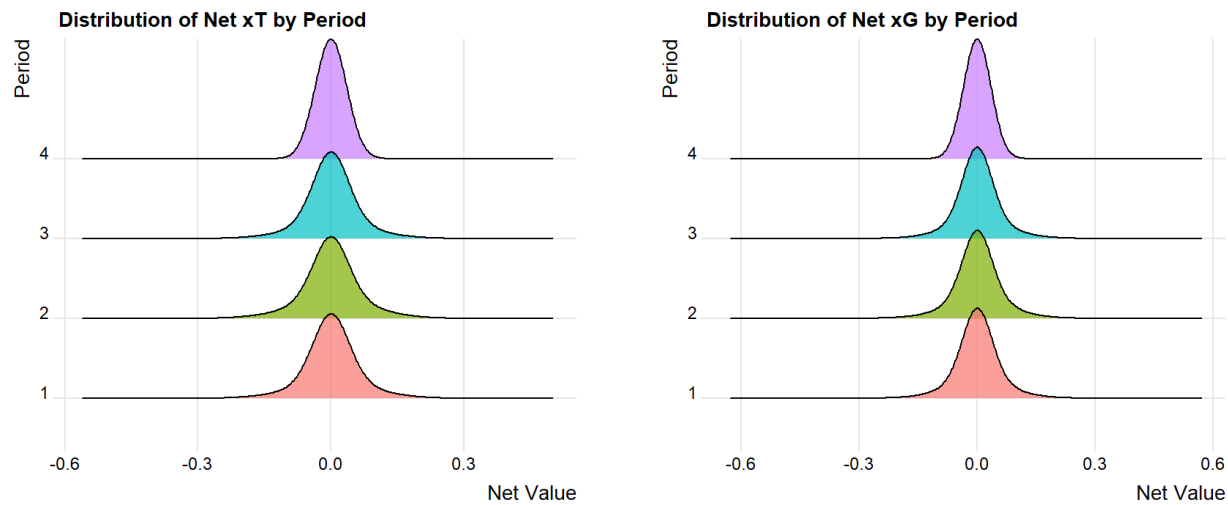

Figure 1: The distributions do not significantly change based on which period the shift occurs. Therefore, it does not seem fit to include a time component in any of our models.

4 Methods

4.1 Overall RAPM - Avery Wang

We built an RAPM model using ridge regression with 10-fold cross validation using both shift-level xT and xG as response variables. The design matrix X follows the usual RAPM convention where, for shift i and player j, $X_{ij} = 1$ if the player is on the ice for the home team and $X_{ij} = -1$ if the player is on the ice for the away team. Hockey rules give the home team a clear advantage by allowing them to put their players on the ice second to optimize matchups, so the model includes an intercept that quantifies this advantage. The response vector y is constructed in the following manner: within shift i, if player j has multiple events assigned to him, we take the average xT (or xG) of those events. If player j has no events, then his xT is 0. Finally, y_i is the net xT for the home team, calculated by the sum of the home players' xT's minus the sum of the away players' xT's. We plan to make a small modification to this y variable to account for shift length. There is one limitation to this model, which is that power plays can cause the number of players on the ice from each team to be uneven. This model naively assumes that, regardless of the number of players on the ice per team, player-level effects remain the same. This is almost certainly not true, so Tyler will attempt to address that in his model.

We plan to either use bootstrapping or a Bayesian approach with normal priors to get uncertainty estimates.

4.2 Special Teams RAPMs - Tyler Quinn

Similarly, we built two RAPM models on special teams datasets using ridge regression with 10-fold cross validation using both shift-level net xT and xG as response variables. The design matrix X follows a different convention than Avery's model described above, by necessity because of the nature of special teams. It is not necessarily true that players preventing scoring chances on the penalty kill also means they will create more scoring chances on the power play. Therefore, we create the design matrix with 2 columns for each player– one to get their power play coefficient, and one to get their penalty kill coefficient. Thus, for player j on shift i, his power play column in X_i is 1 if he is on the power play. If he is on the penalty kill, then his penalty kill column is -1. Therefore, we lose information about home ice advantage on special teams, but the intercept term under this model is the average advantage in net xT or xG experienced by the team on the power play.

We also plan to use bootstrapping or a Bayesian approach with Normally distributed priors to get estimates of uncertainty in these coefficients.

4.3 Promotion to NHL - Alex Hill

We are building a Logistic Regression model with a binary indicator for if a player made it to the NHL as the response variable, xT as the main covariate of interest, and other predictors including age and position for additional context. After getting player coefficients for xT and analyzing how odd-man situations factor into the equation, we want to get an idea of AHL xT is at predicting whether a player makes it to the NHL. Further, we can get interesting insights from this model regarding xT's importance across different player groups. For instance, we would expect that xT for forwards, and possibly wingers more specifically, would be more predictive of NHL promotion since those positions are obviously more centered around goal-scoring and chance creation. It's also possible that younger players with high xT have a greater probability of promotion to the NHL if those players are truly dynamic scorers who can make an impact at the AHL (and NHL) level at such a young age.

To evaluate the model, we can look at calibration plots to compare NHL promotion probabilities to actual proportions of players who made it to the NHL with those probabilities bins. We can also split this across different values of xT in the AHL, position groups, age groups, and other included covariates to get a more in-depth look at how the model performs. Additionally, while not specific to logistic regression, we could run a basic correlation analysis on xT values with basic scoring statistics like points (goals and assists) or plus-minus to investigate the stability of xT and how it correlates to scoring at the NHL level.

Model uncertainty will naturally come from coefficient standard errors. However, we could use bootstrapping to generate hypothetical player xT-Age-Position combinations to increase our player sample size and understanding of the importance of each covariate in predicting NHL promotion. We could also use cross validation to randomly assign different groups of players to folds and get a more holistic and thorough idea of our model's performance.

5 Results

5.1 Overall RAPM - Avery Wang

The players with the top ten highest RAPM coefficients are shown in Table 1 and Table 2. We may eventually replace this table with a graph of distributions to display uncertainty about these estimates. We will fill in the third column later and may change it to say number of career games played, or a similar metric that demonstrates prolonged NHL success. As expected, the coefficients from the two models follow a strong linear relationship, as shown in Figure 2. The models estimate that home-ice advantage is worth 7.66×10^{-4} on the xT scale and 4.68×10^{-4} on the xG scale. Multiplying the xT home-ice value by the conversion factor of 0.3695 from Figure 2 gives a home-ice advantage of 2.83×10^{-4} on the xG scale. This means that the xG model estimates that home-ice advantage is small, but 65% stronger than the estimate from the xT model.

Player	Coefficient	Played in NHL?
Will Lockwood	4.13×10^{-3}	
Xavier Simoneau	3.35×10^{-3}	
Michael Mersch	3.13×10^{-3}	
Hardy Haman Aktell	3.07×10^{-3}	
Vinnie Hinostroza	2.91×10^{-3}	
Marian Studenic	2.75×10^{-3}	
Nic Petan	2.65×10^{-3}	
Cameron Morrison	2.52×10^{-3}	
Vasily Podkolzin	2.49×10^{-3}	
Milos Kelemin	2.42×10^{-3}	

Table 1: The players with the highest RAPM coefficients for the xT model.

Player	Coefficient	Played in NHL?
Dustin Wolf	1.76×10^{-3}	
Wyatte Wylie	1.61×10^{-3}	
Nic Petan	1.54×10^{-3}	
Alex Belzile	1.53×10^{-3}	
Hardy Haman Aktell	1.49×10^{-3}	
Max Crozier	1.39×10^{-3}	
Will Lockwood	1.38×10^{-3}	
Joe Snively	1.31×10^{-3}	
Wade Allison	1.29×10^{-3}	
Leon Gawanke	1.28×10^{-3}	

Table 2: The players with the highest RAPM coefficients for the xG model.

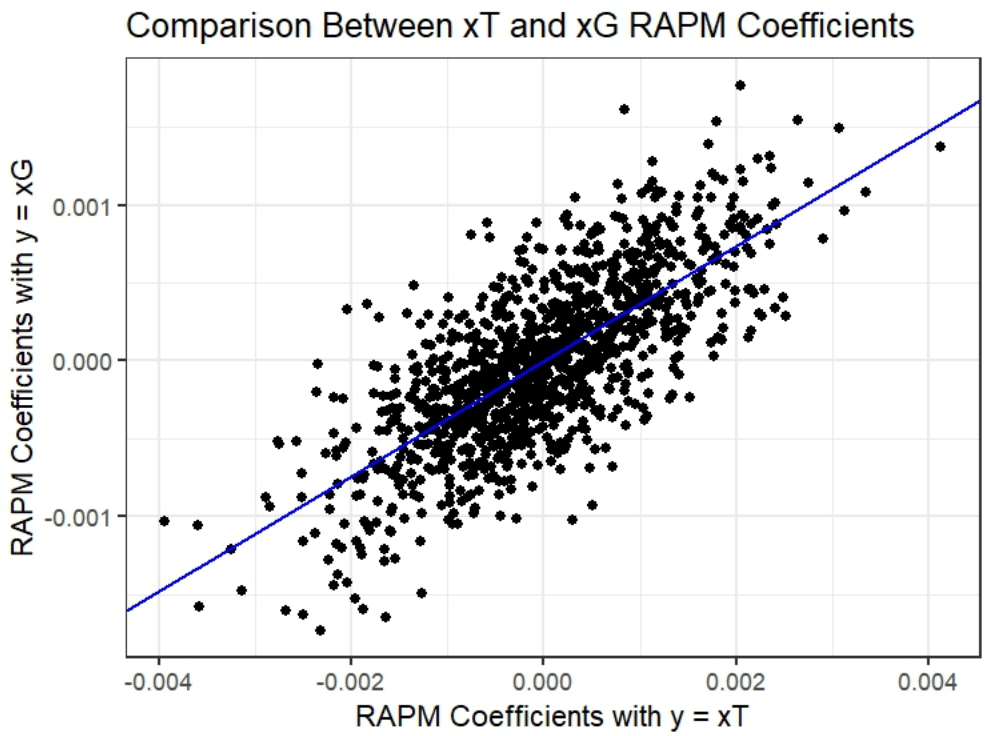

Figure 2: There is a strong linear relationship between the players' RAPM coefficients when using xT and xG as the model's response variable. The blue line shows the zero-intercept fit using ordinary least squares, and is given by the equation $y = 0.3695x$, $R^2 = 0.5428$.

5.2 Special Teams RAPMs - Tyler Quinn

The model has not been fit yet, so there are no relevant results at this time.

6 Discussion

Hopefully, we see that higher xT and xG RAPM coefficients lead to a higher probability of NHL appearance or more career games played.

6.1 Limitations and Future Work

This project does have some limitations in scope and interpretation. Notably, any flaws in the underlying xT or xG translate to this work. Some problems are addressed by dividing credit using RAPMs, but if they are systematically biased toward a certain type of player, those benefits or deductions are carried over. Additionally, players in hockey are deployed in lines, and these lines are often the same throughout large parts of the season. This leads to heavy multicollinearity in the data. Ridge regression and the RAPM approach divides credit evenly for players who are almost always on the ice together. We think this is a reasonable approach but it may not accurately describe the situation if two players usually play together but one of them is the main driver of their results. Future work might use more granular

data that describes exactly which players are involved in events, or infer it using continuous tracking data, to gain more accurate results in these cases.

Using a logistic regression model is also not optimal for measuring NHL performance. This method treats players the same whether they are superstars or merely play one shift at the NHL level. Future work might regress on some continuous measure of NHL success for more accurate results.

7 References

Sezgin, M., & Quinn, T. (2026). Introducing a Gaussian Rink Control-Based Expected Threat Framework to Hockey. *HALO 2026 Competition*.

8 Appendix

All code can be found in this GitHub repository.